SOMAIYA VIDYAVIHAR UNIVERSITY
K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

K. J. Somaiya School of Engineering, Mumbai
Department of Computer Engineering · Mini Project, Semester VII, 2026-2027

Somaiya TRUST

Element

Combine anything. Discover what the world hasn't named yet.
An iOS crafting game that runs its AI entirely on the phone.

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~2,700	16	125	2	0
lines of Swift	source files	offline recipes	game worlds	network calls or packages

The idea

Little Alchemy uses a fixed list of recipes. Infinite Craft lets a language model decide what two things make. Element asks whether that can work with no server at all.

The player drags two elements together. If the pair is new, Apple's on-device **Foundation Models** framework invents the result. It needs no API key and no connection, and the player's combinations stay on the phone.

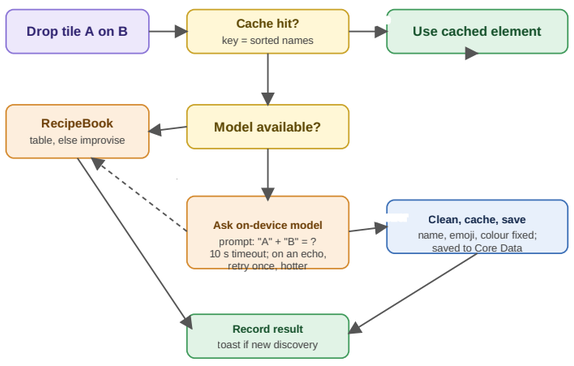
Objectives

- A drag and drop crafting canvas in SwiftUI
- Structured output from the on-device model
- A combination that never ends in an error
- Saved discoveries and a combination cache
- Two worlds with separate progress
- A searchable log with JSON export, and a map

Features

- Tap a tile, or press and hold to drag it out of the tray
- Drop one tile on another to combine; double-tap to duplicate
- Drag a tile back to the tray to remove it
- Searchable, sortable tray and a favourites filter
- Recipes for each element, JSON export of the log
- Atlas map with a stable pin for every element
- Light, dark and reduce motion settings

How a pair is resolved



Cache first, then the model, then the recipe book

Guided generation

A `@Generable` struct makes the model return a name, an emoji, a colour and a description in a fixed shape.

The app still treats the output as untrusted. It strips stray punctuation from names, keeps one real emoji, forces a valid six digit hex colour, and reuses an existing element when the name matches.

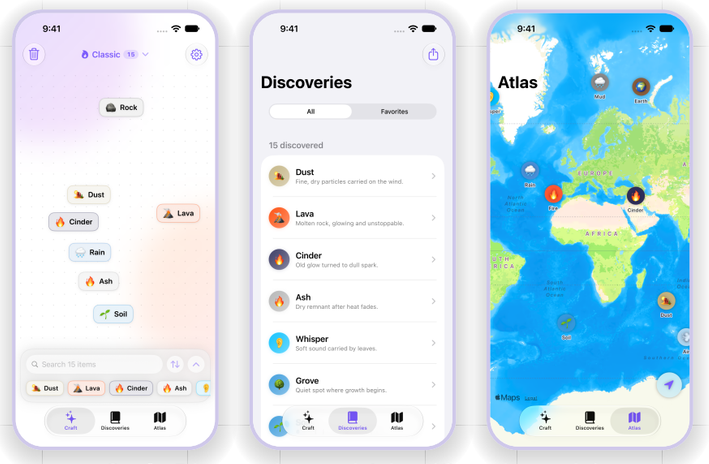
When things go wrong

- Model unavailable: a banner explains why, and the recipe book answers
- No answer in 10 seconds: request cancelled, recipe book answers
- Model repeats an input: one retry, hotter, then fallback
- Fallback results are not cached, so the model can try again later

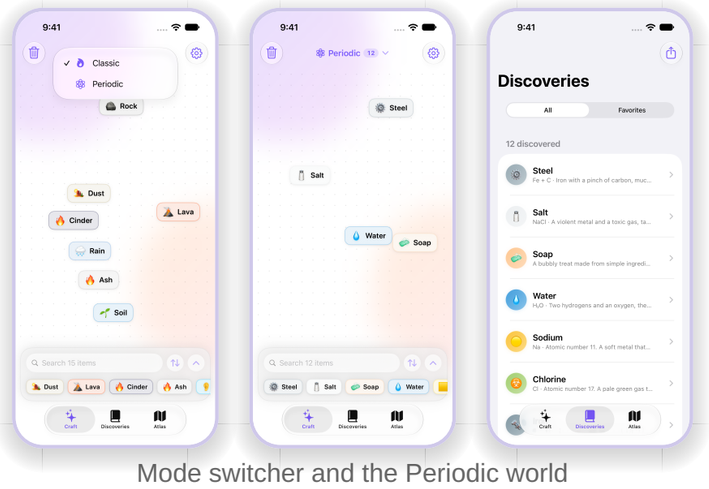
Two worlds

	Classic	Periodic
Start	Fire, Water, Earth, Air	8 real elements
Example	Water + Earth = Plant	Sodium + Chlorine = Salt
Prompt	Most obvious result	Real compound or material
Offline	93 recipes	32 recipes

Screens

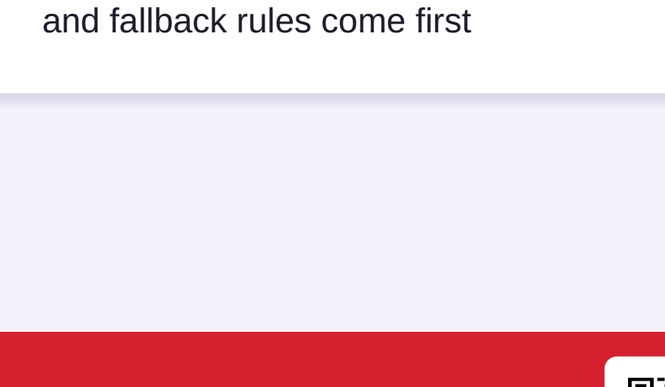


Craft, Discoveries and Atlas (Classic mode)



Mode switcher and the Periodic world

Architecture



One `@Observable` view model between the views and three data sources

Limits and next steps

- Needs iOS 26.5 and Apple Intelligence for generated results
- Answer quality was not measured
- No automated tests yet. The cache key and fallback rules come first

Built with SwiftUI · Foundation Models · Core Data · MapKit · Observation

Source code and report: github.com/altf4-games/Element